



Dr. Prateek Raj Gautam

Ph.D., Electronics and Communication Engineering,
Motilal Nehru National Institute of Technology Allahabad

Email : dr.prateekrajgautam@gmail.com | prateek@mgeek.in

Webpage : <https://prateekrajgautam.github.io>

Github : <https://github.com/prateekrajgautam>

Mobile : +91 - 9151 404 899

ORCID:0000-0002-2889-4275,

WOS:1552109,

IEEE:91250146,

SCHOLAR:slZHj6cAAAAJ

PUBLICATIONS JOURNAL(J) CONFERENCE(C)

- [J1] **P. R. Gautam**, S. Kumar, A. Verma, T. Rashid, et al., "Energy-efficient localization of sensor nodes in WSNs using beacons from rotating directional antenna," *IEEE Transactions on Industrial Informatics*, vol. 15, no. 11, pp. 5827–5836, Nov. 2019. DOI: [10.1109/tii.2019.2908437](https://doi.org/10.1109/tii.2019.2908437) issn 1551-3203 **Impact Factor: 12.3** **SCIE, Q1**
- [J2] **P. R. Gautam**, S. Kumar, A. Verma, and A. Kumar, "Energy-efficient localization of sensor nodes in wsns using single beacon node," *IET Communications*, vol. 14, no. 9, pp. 1459–1466, 2020. DOI: [10.1049/iet-com.2019.1298](https://doi.org/10.1049/iet-com.2019.1298) issn 1751-8628 **Impact Factor: 1.6** **SCIE, Q2**
- [J3] A. Verma, S. Kumar, **P. R. Gautam**, and A. Kumar, "Fuzzy logic based effective clustering of homogeneous wireless sensor networks for mobile sink," *IEEE Sensors Journal*, vol. 20, no. 10, pp. 5615–5623, May 2020. DOI: [10.1109/jsen.2020.2969697](https://doi.org/10.1109/jsen.2020.2969697) issn 1530-437X **Impact Factor: 4.3** **SCIE, Q1**
- [J4] A. Verma, S. Kumar, **P. R. Gautam**, and A. Kumar, "Neural-fuzzy based effective clustering for large-scale wireless sensor networks with mobile sink," *Peer-to-Peer Networking and Applications*, Jun. 2021. DOI: [10.1007/s12083-021-01167-6](https://doi.org/10.1007/s12083-021-01167-6) issn 1936-6450 **Impact Factor: 4.2** **SCIE, Q2**
- [J5] A. Verma, S. Kumar, **P. R. Gautam**, T. Rashid, et al., "Broadcast and reliable coverage based efficient recursive routing in large-scale wsns," *Telecommunication Systems*, vol. 75, no. 1, pp. 63–78, Jun. 2020. DOI: [10.1007/s11235-020-00679-5](https://doi.org/10.1007/s11235-020-00679-5) issn 1572-9451 **Impact Factor: 2.5** **SCIE, Q2**
- [J6] M. Yadav, **P. R. Gautam**, V. Shokeen, and P. K. Singhal, "Modern fisher-yates shuffling based random interleaver design for SCFDMA-IDMA systems," *Wireless Personal Communications*, vol. 97, no. 1, pp. 63–73, May 2017. DOI: [10.1007/s11277-017-4492-9](https://doi.org/10.1007/s11277-017-4492-9) issn 0929-6212 **Impact Factor: 2.2** **SCIE, Q2**
- [J7] A. Verma, T. Rashid, **P. R. Gautam**, S. Kumar, et al., "Cost and sub-epoch based stable energy-efficient clustering algorithm for heterogeneous wireless sensor networks," *Wireless Personal Communications*, vol. 107, no. 4, pp. 1865–1879, Apr. 2019. DOI: [10.1007/s11277-019-06362-6](https://doi.org/10.1007/s11277-019-06362-6) issn 0929-6212 **Impact Factor: 2.2** **SCIE, Q2**
- [J8] T. Rashid, S. Kumar, A. Verma, **P. R. Gautam**, et al., "Co-reerp: Cooperative reliable and energy efficient routing protocol for intra body sensor network (intra-wbsn)," *Wireless Personal Communications*, vol. 114, no. 2, pp. 927–948, Apr. 2020. DOI: [10.1007/s11277-020-07401-3](https://doi.org/10.1007/s11277-020-07401-3) issn 0929-6212 **Impact Factor: 2.2** **SCIE, Q2**
- [J9] S. Kumar, **P. R. Gautam**, A. Verma, T. Rashid, et al., "An energy-efficient transmission in wsns for different climatic conditions," *Wireless Personal Communications*, vol. 110, no. 1, pp. 423–444, Sep. 2019. DOI: [10.1007/s11277-019-06735-x](https://doi.org/10.1007/s11277-019-06735-x) issn 0929-6212 **Impact Factor: 2.2** **SCIE, Q2**
- [J10] S. Kumar, **P. R. Gautam**, T. Rashid, A. Verma, et al., "Division algorithm based energy-efficient routing in wireless sensor networks," *Wireless Personal Communications*, Aug. 2021. DOI: [10.1007/s11277-021-08996-x](https://doi.org/10.1007/s11277-021-08996-x) issn 1572-834X **Impact Factor: 2.2** **SCIE, Q2**
- [J11] R. C. S. Chauhan, A. Kumar, and **P. R. Gautam**, "Optical orthogonal code generation scheme and grouping of codes for optical CDMA systems," *International Journal of System Assurance Engineering and Management*, vol. 12, no. 1, pp. 91–103, 1 Jun. 2020. DOI: [10.1007/s13198-020-01007-5](https://doi.org/10.1007/s13198-020-01007-5) issn 0976-4348 **Impact Factor: 2** **SCIE, Q3**
- [J12] **P. R. Gautam**, A. Verma, S. Kumar, D. Prasad, et al., "Design of directional antennas for wireless sensor networks and the internet of things experiments," *IEEE Sensors Letters*, vol. 6, no. 9, pp. 1–4, 2022. DOI: [10.1109/LSSENS.2022.3202919](https://doi.org/10.1109/LSSENS.2022.3202919) issn 2475-1472 **Impact Factor: 2.8** **SCIE, Q2**
- [J13] Shilpi, **P. R. Gautam**, S. Kumar, and A. Kumar, "An optimized sensor node localization approach for wireless sensor networks using rssi," *The Journal of Supercomputing*, vol. 79, pp. 7692–7716, 2022. DOI: <https://doi.org/10.1007/s11227-022-04971-w> issn 0920-8542 **Impact Factor: 3.3** **SCIE, Q2**

- [J14] A. Verma, S. Kumar, **P. R. Gautam**, T. Rashid, et al., “Enhanced cost and sub-epoch based stable energy-efficient clustering algorithm for heterogeneous wireless sensor networks,” *Wireless Personal Communications*, Jul. 2023. DOI: [10.1007/s11277-023-10601-2](https://doi.org/10.1007/s11277-023-10601-2) issn [1572-834X](#) **Impact Factor: 2.2** **SCIE, Q2**
- [J15] S. Kumar, **P. R. Gautam**, T. Rashid, A. Verma, et al., “ETDCC: Energy-efficient transmission scheme for dynamic climatic conditions in WSN,” *TELKOMNIKA (Telecommunication Computing Electronics and Control)*, vol. 16, no. 3, p. 1126, Jun. 2018. DOI: [10.12928/telkomnika.v16i3.8513](https://doi.org/10.12928/telkomnika.v16i3.8513) issn [1693-6930](#) **Scopus**
- [J16] T. Rashid, S. Kumar, A. Verma, **P. R. Gautam**, et al., “Pm-EEMRP: Postural movement based energy efficient multi-hop routing protocol for intra wireless body sensor network (intra-WBSN),” *TELKOMNIKA (Telecommunication Computing Electronics and Control)*, vol. 16, no. 1, p. 166, Feb. 2018. DOI: [10.12928/telkomnika.v16i1.7318](https://doi.org/10.12928/telkomnika.v16i1.7318) issn [1693-6930](#) **Scopus**
- [J17] A. Verma, T. Rashid, **P. R. Gautam**, S. Kumar, et al., “Fuzzy based stable clustering protocol for Heterogeneous wireless sensor networks,” *International Journal of Engineering and Technology*, vol. 9, no. 4, pp. 2854–2860, Aug. 2017. DOI: [10.21817/ijet/2017/v9i4/170904046](https://doi.org/10.21817/ijet/2017/v9i4/170904046) issn [0975-4024](#) **Scopus 2017**
- [J18] T. Rashid, S. Kumar, A. Verma, **P. R. Gautam**, et al., “RB-IEMRP: Relay based improved throughput energy-efficient multi-hop routing protocol for intra body sensor network (INTRA-WBSN),” *International Journal of Computer Networks & Communications*, vol. 11, no. 02, pp. 69–82, Mar. 2019. DOI: [10.5121/ijcnc.2019.11205](https://doi.org/10.5121/ijcnc.2019.11205) issn [0974-9322](#) **Scopus**
- [C1] **P. R. Gautam**, S. Kumar, A. Verma, and A. Kumar, “Localization of sensor nodes in WSNs using three dimensional angle of arrival detection at BS,” in *2019 International Conference on Electrical, Electronics and Computer Engineering (UPCON)*, ZHCET, AMU, Aligarh: IEEE, Nov. 2019, pp. 1–4. DOI: [10.1109/upcon47278.2019.8980262](https://doi.org/10.1109/upcon47278.2019.8980262) isbn: 9781728134550 issn [2687-7767](#) **Scopus**
- [C2] **P. R. Gautam**, S. Kumar, A. Verma, T. Rashid, et al., *Localization of Sensor Nodes in WSN Using Area Between a Node and Two Beacons* (Lecture Notes in Electrical Engineering). Motilal Nehru National Institute of Technology, Allahabad: Springer, Dec. 2019, vol. 587, pp. 221–228, 1060 pp. DOI: [10.1007/978-981-32-9775-3_22](https://doi.org/10.1007/978-981-32-9775-3_22) isbn: 9813297743 issn [1876-1100](#) **Book chapter**
- [C3] **P. R. Gautam**, S. Kumar, and A. Kumar, “Sensor localization in wsns using rotating directional - antenna at the base station,” in *Advances in VLSI, Communication, and Signal Processing*, ser. Lecture Notes in Electrical Engineering, vol. 683, Motilal Nehru National Institute of Technology, Allahabad: Springer, Oct. 2020, pp. 705–718. DOI: [10.1007/978-981-15-6840-4_58](https://doi.org/10.1007/978-981-15-6840-4_58) isbn: 978-981-15-6839-8 issn [1876-1100](#) **Book chapter**
- [C4] A. Kumar, S. Kumar, **P. R. Gautam**, A. Verma, et al., *Performance Evaluation of Multi-operands Floating-Point Adder* (Lecture Notes in Electrical Engineering). JK Institute of Applied Physics and Technology, Allahabad University, Allahabad: Springer Singapore, Dec. 2019, vol. 524, pp. 537–546. DOI: [10.1007/978-981-13-2685-1_51](https://doi.org/10.1007/978-981-13-2685-1_51) isbn: 9811326843 issn [1876-1119](#) **Book chapter**
- [C5] S. Kumar, A. Verma, **P. R. Gautam**, A. Dayal, et al., “The load balancing of optimizing LEACH clustering algorithm with mobile sink and rendezvous nodes,” in *2018 5th IEEE Uttar Pradesh Section International Conference on Electrical, Electronics and Computer Engineering (UPCON)*, Madan Mohan Malaviya University of Technology, Gorakhpur: IEEE, Nov. 2018. DOI: [10.1109/upcon.2018.8596989](https://doi.org/10.1109/upcon.2018.8596989) isbn: 978-1-5386-5002-8 issn [2687-7759](#) **Scopus**
- [C6] S. Kumar, **P. R. Gautam**, A. Verma, R. Verma, et al., *Energy Efficient Routing using Sectors Based Energy-Hole Reduction in WSNs*. ZHCET, AMU, Aligarh: IEEE, 2019. DOI: [10.1109/upcon47278.2019.8980254](https://doi.org/10.1109/upcon47278.2019.8980254) isbn: 978-1-7281-3455-0 issn [2687-7767](#) **Scopus**
- [C7] A. Verma, S. Kumar, **P. R. Gautam**, and A. Kumar, *Stable Energy-Efficient Routing Algorithm for Dynamic Heterogeneous Wireless Sensor Networks* (Lecture Notes in Electrical Engineering). Motilal Nehru National Institute of Technology, Allahabad: Springer, Dec. 2019, vol. 587, pp. 221–228, 1060 pp. DOI: [10.1007/978-981-32-9775-3_15](https://doi.org/10.1007/978-981-32-9775-3_15) isbn: 9813297743 issn [1876-1100](#) **Book chapter**
- [C8] S. Shilpi, **P. R. Gautam**, S. Kumar, and A. Kumar, “A comparative analysis of distance-based node localization in wireless sensor network,” in *2021 8th International Conference on Signal Processing and Integrated Networks (SPIN)*, vol. 0, 2021, pp. 118–123. DOI: [10.1109/SPIN52536.2021.9566136](https://doi.org/10.1109/SPIN52536.2021.9566136) isbn: 9781665435642 issn [0](#) **Scopus**

- [C9] M. Yadav, **P. R. Gautam**, and K. Singhal P., “Inverse tree interleavers in uav communications for interference mitigation,” in *Decision Support Systems for Smart City Applications* (Concise Introductions to AI and Data Science), Concise Introductions to AI and Data Science. John Wiley & Sons, Ltd, Dec. 2022, ch. 3, pp. 35–52. DOI: [10.1002/9781119896951.ch3](https://doi.org/10.1002/9781119896951.ch3) isbn: 9781119896951 issn: [scopus](https://scopus.com) **Book chapter**
- [C10] A. Rukasar and P. R. Gautam, “Lane detection and tracking algorithms for driver assistance system,” in 2023, pp. 872–879. DOI: [10.1109/icac3n60023.2023.10541447](https://doi.org/10.1109/icac3n60023.2023.10541447) isbn: 9798350330861 issn: [scopus](https://scopus.com)
- [C11] V. Kumar, A. Kumar, and **P. R. Gautam**, “Dental disease detection and classification in radiograph images using deep learning model,” in 2023, pp. 1198–1203. DOI: [10.1109/ICAC3N60023.2023.10541747](https://doi.org/10.1109/ICAC3N60023.2023.10541747) isbn: 9798350330861 issn: [scopus](https://scopus.com)
- [C12] N. Awasthi, **P. R. Gautam**, and A. Sharma, “Rfecv-dt: Recursive feature selection with cross validation using decision tree based android malware detection,” in 2024. DOI: [10.1109/ICCCNT61001.2024.10725127](https://doi.org/10.1109/ICCCNT61001.2024.10725127) isbn: 9798350370249 issn: [scopus](https://scopus.com)
- [C13] M. Ansari and **P. R. Gautam**, “Classification of soil moisture content with the application of deep learning,” in 2024. DOI: [10.1109/ICIC3S61846.2024.10603387](https://doi.org/10.1109/ICIC3S61846.2024.10603387) isbn: 9798350364088 issn: [scopus](https://scopus.com)
- [C14] A. Sharma, A. Upadhyay, and **P. R. Gautam**, “Prediction of water discharge in mahanadi river basin, india using artificial neural networks,” in 2025, vol. 1, pp. 28–32. DOI: [10.1201/9781003501244-5](https://doi.org/10.1201/9781003501244-5) isbn: 9781032911571 issn: [scopus](https://scopus.com)
- [C15] P. Mishra, J. Singh, and **P. R. Gautam**, “Mustard and wheat mildew disease classification using deep learning,” in 2025, vol. 1, pp. 72–78. DOI: [10.1201/9781003501244-14](https://doi.org/10.1201/9781003501244-14) isbn: 9781032911571 issn: [scopus](https://scopus.com)
- [C16] A. Upadhyay, A. Sharma, and **P. R. Gautam**, “Estimation of sediment load in mahanadi river, india using artificial neural networks,” in 2025, vol. 1, pp. 23–27. DOI: [10.1201/9781003501244-4](https://doi.org/10.1201/9781003501244-4) isbn: 9781032911571 issn: [scopus](https://scopus.com)
- [C17] M. Ansari and **P. R. Gautam**, “Comparison of different pre-trained deep learning models for classification of soil moisture content,” in 2025, vol. 2, pp. 107–113. DOI: [10.1201/9781003561651-15](https://doi.org/10.1201/9781003561651-15) isbn: 9781032911571 issn: [scopus](https://scopus.com)

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“**Sensor Localization in WSNs Using Rotating Directional - Antenna at the Base Station.**” In *2nd International Conference on VLSI, Communication and Signal Processing (VCAS 2019)* at MNNIT Allahabad (UP) India. 21st – 23rd October 2019
